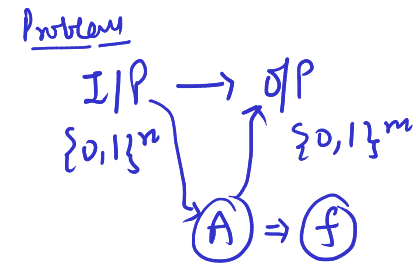


Computational Problems

$$f : \{0, 1\}^n \longrightarrow \{0, 1\}^m$$



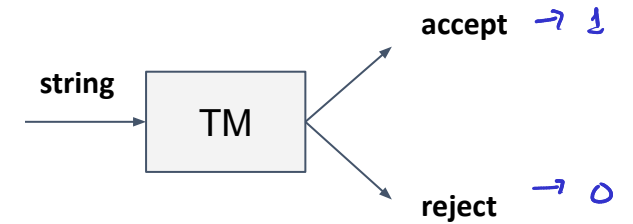
$$f(I/P) \rightarrow O/P$$

Decision Problems

- The problem can be answered in 'YES' or 'NO'
 - Boolean Formula Satisfiability (**SAT**)
 - Given a boolean formula in conjunctive normal form, is there an assignment that satisfies the formula?

$f(x_1, x_2, x_3, x_4)$
 (2^4)
 $\begin{matrix} 0000 \\ \vdots \\ 1111 \end{matrix}$

$$f = (x_1 \vee \overline{x_2}) \wedge (x_1 \vee x_3 \vee \overline{x_4}) \wedge (\overline{x_2} \vee \overline{x_3}) \wedge (\overline{x_4}) \quad \text{is it?}$$



- Decision problems can be represented as
 - Acceptance of string over a language
 - Direct correspondence between a problem and its language

$$L : \{0, 1\}^* \rightarrow \{0, 1\}$$

(Handwritten note: 'yes' for 1, 'no' for 0)

$$L(x) = \begin{cases} 1 & \text{if } x \in L \\ 0 & \text{if } x \notin L \end{cases}$$

(Handwritten note: arrow from L(x) to L in the denominator)

Computational Problems

✓ Search Problems *(explore a search space)*

- Find the instance(s) that satisfies the search criteria
 - **FIND-SAT**: Given a boolean formula in conjunctive normal form, find an assignment that satisfies it if such an assignment exists.

✓ $(x_1 \vee \bar{x}_2) \wedge (x_1 \vee x_3 \vee \bar{x}_4) \wedge (\bar{x}_2 \vee \bar{x}_3) \wedge (\bar{x}_4)$

$x_1 = 1, x_2 = 0, x_3 = 1, x_4 = 0$

• Optimization Problems

- Find the instance(s) that optimize (maximize or minimize) an objective function under a set of constraints on the problem variables
 - **MAX-SAT**: Given a boolean formula in conjunctive normal form, find an assignment that satisfies as many of its clauses as possible.

Combs. of x_1, x_2, x_3, x_4

$$\underbrace{(x_1 \vee \bar{x}_2)}_{k_1} \wedge \underbrace{(x_1 \vee x_3 \vee \bar{x}_4)}_{k_2} \wedge \underbrace{(\bar{x}_2 \vee \bar{x}_3)}_{k_3} \wedge \underbrace{(\bar{x}_4)}_{k_4}$$

✓ $\max \sum k_i$

$k_i = \begin{cases} 1 & \text{if clause } i \text{ is satisfied} \\ 0 & \text{if clause } i \text{ is not satisfied} \end{cases}$

$x_i = \{0, 1\}$

Computational Problems

- Transformation of Optimization Problem into Decision Problem

- Optimization problems can be re-casted into decision problems
 - SHORTEST-PATH: Given an unweighted, undirected graph and a pair of nodes, find the shortest path between them.

$$P(\{G(V, E), \{v_s, v_e\}\}, \{p\})$$

$$\{p\} = v_s, \underline{v_1, v_2, \dots, v_n}, v_e$$

- DEC-SHORTEST-PATH: Does there exist a path whose length is at-most k ?

$$P(\{G(V, E), \{v_s, v_e\}, \underline{k}\}, \{0, 1\})$$

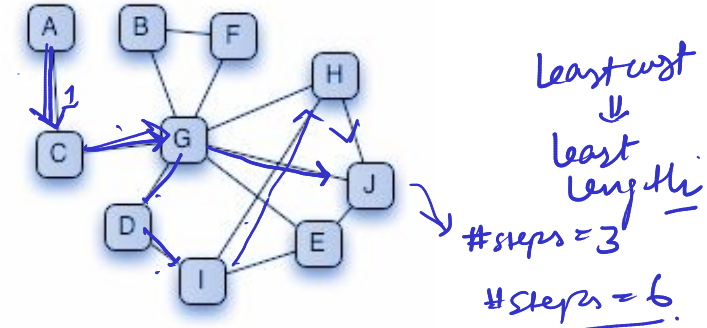
$$\{p\} = v_s, v_1, v_2, \dots, v_n, v_e \checkmark$$

$$\exists \{p\} s.t. C(p) \leq k ? \rightarrow \text{bound}$$

$$\checkmark C(p) = C(v_s, v_1) + c(v_1, v_2) + \dots + c(v_n, v_e)$$

- ✓→ **Hard** decision version implies **hard** optimization version.
- ✓→ **Easy** optimization version implies **easy** decision version.
- Complexity theory is mainly studied in the context of decision problems.

$$\rightarrow \underline{Q(Y, E)} \checkmark \quad \underline{\{A, J\}}$$



Exercise:

1. Learn about TSP
2. Create the decision version of TSP

